NIHVA

NIHVA Technologies Pvt. Ltd.

Sliding Doors



NIHVA Hanger Sliding Doors are manufactured by state-of-art & unique fabrication techniques which would allow sturdy frame and minimum deflection. Number of door leafs are decided on the basis of clear opening required and the space available. Door parting can be designed to be uni-directional or bi-directional as per customer and site requirement. Each door leaf is operated by its own independent driving unit, capacity of which is defined by the door load it has to drive. Entire weight of the door is carried by bottom rolling wheels with high load bearing arran gement for its smooth operation. Guiding rollers are provided on the upper side of doors to guide them throughout the travel span. Firm movement of doors are achieved by flawless design and heavy duty upper and lower track mechanisms.

Weather seal systems are designed to provide all weather protection as well as to accommodate all deflections in building and door to ensure proper sealing. Rusty environmental conditions are prevailed with the protective epoxy primer coats and polyurethane paint.

Our robust and flawless design will therefore help you perform your activities fluidly.

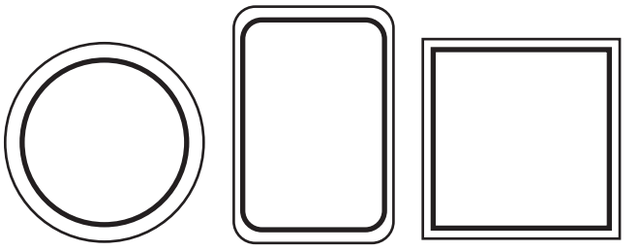


Leading Attributes

- Manufactured using hot-rolled structural steels and unique fabrication techniques.
- Design to withstand high wind-pressure which is ensured in Stad-pro software.
- Height of up to 35-40 meters and width as per requirement.
- Telescopic type top roller cum guiding system to accommodate the deflections.
- Galvanized skirting to provide aesthetic look and to cover the driving mechanisms.
- All type weather sealing with aluminum extrusion base.
- Wide range of cladding design and color.
- Heavy duty buffer stops on extreme ends for each door.
- Independent driving mechanism for each door movement .
- High efficiency driving mechanism with brake-motor which accords flawless operation and immediate braking system.
- Aesthetic Closure for all mechanism for protection and easy maintenance.
- Efficient Towing arrangement for manual operation.
- User friendly control panels which will accord with integrated operation and all safety measures.
- Easy construction and assembly design for site erection.
- Integrated safety devices to eliminate the possible accidents.

Dimensions

As standard the following window sizes are available using 8/9mm safety glass. On request special glass with low U value, Georgian wired, double glazed can be used to ensure building control compliance.



ø 530mm	**700 x 700mm**
500 x 600mm	**700 x 1,000mm**
500 x 800mm	**700 x 1,500mm**
580 x 880mm	**700 x 2,000mm**



Application	:	External Internal
Size range		
Max. Width	:	7 to 8 mtr
Max. Height (subject to application)	:	15 to 20 mtr (subject to application)
Arrangement	:	Max 20+20 configuration
Operation	:	Standard (Single Drive) Power (Independent Drive)
Speed (dependant on size)	:	30 mtr per sec with either 240v single phase or 415v three phase, earth and neutral motor and traction chain drive.
Emergency Opening	:	Standard
Curtain	:	55mm, 65mm & 80mm thick high density foam filled twin skin steel panel with concealed hinge system are used subject to height of door
Wind load resistance	:	EN12424 – Class 4 – 140km/h
Mechanical Compliance	:	BS EN12604
Safe Use Of Power Operated Doors	:	BS EN12453
Safety Equipment	:	DIN EN13241 – 1

Bottom Tracks

Bottom Tracks are made up of flat bottom rail tracks which are mounted on cross tying members. Cross members are provided with leveling bolts to ensure the zero water level of the track throughout its span. This arrangement is then allowed to submerge in the civil work leaving only the wheel track surfaced.



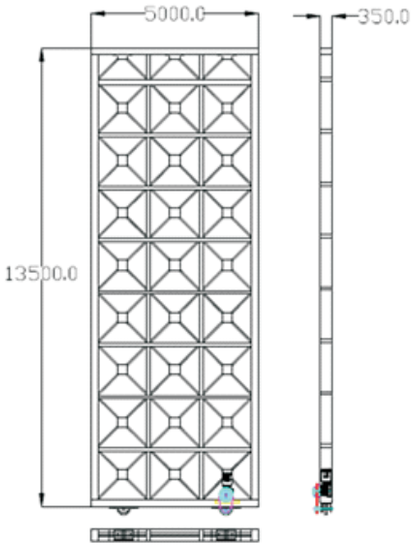


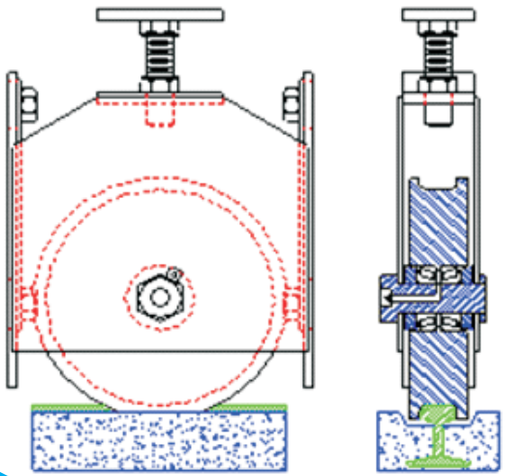
Top Rails

Top rails are made up of wide flange I-Beams which are mounted directly on the structure of facility/Shop. Wide Flange accommodates telescopic roller arrangement effectively which enables the stable movement of door leafs. Weight of the top rails are limited to 40kg/meter

Leaf Frame

Leaf Frame are fabricated from hot-rolled structural steel and bolted together with high strength friction grip bolts. Main frame structure is supported by I sectional cross members and long members. Whole structure is then tied together using high tension sustaining tying ropes. Door Leafs can be fabricated up to 15m height and 7m of width per leaf.



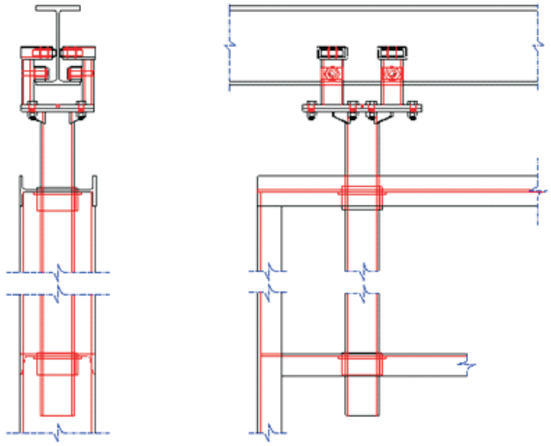


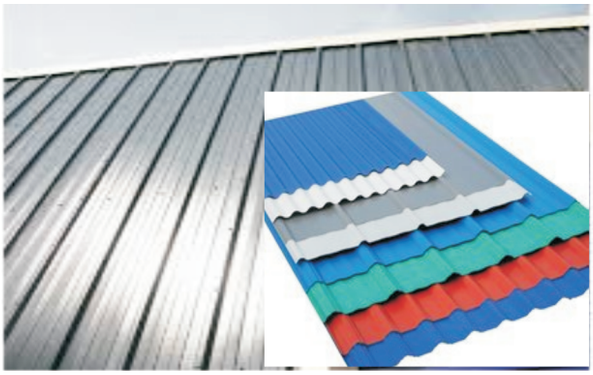
Bottom Rolling

Door leafs are assembled at bottom with double flanged hardened steel wheels, coupled with high load bearings for frictionless operation. Number and size of wheels can be decided depending on the length of door leaf and the load it has to bear. These wheels are capable to withstand dead loads as well as wind loads hence providing greater stability to the door leafs. Adjustable jack plate arrangement allows the wheel to be adjusted down for maximum traction or adjusted up off the floor in case of power failure.

Top Roller Guiding

Top roller guider are used to guide the door leaf throughout its travel and to maintain stability of the door. Top roller guiding can be fixed type and telescopic type depending upon the vertical deflection of the building structure. Fixed type roller guides can be used for door leaf of maximum 5m height and with negligible vertical deflection in building structure. Telescopic roller guiding mechanism is used for door leafs exceeding the height of 5m and are capable of accommodating up to 100mm of vertical deflection of building structure. A trolley head of telescopic guiding mechanism is free to fall and rise vertically through a low friction bearing mechanism which allows smooth operation of the door during its travel.



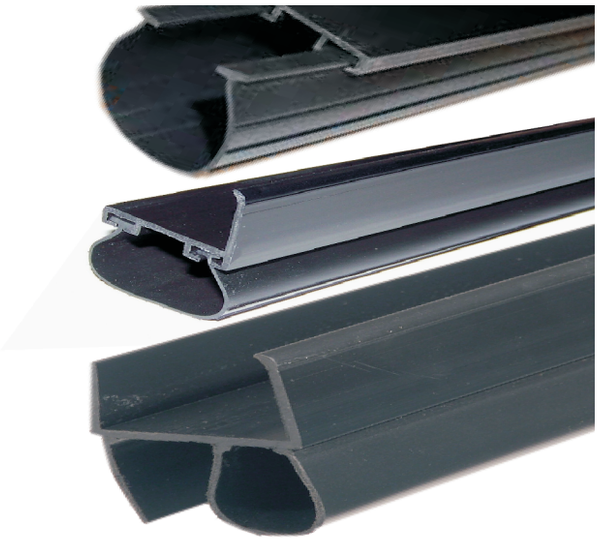


Door Cladding

Cladding enclosure is provided on the entire door leaf, thickness of which is defined by the wind load it has to withstand. Cladding can be either single side or both side depending on the customer requirement. Cladding also can be in different shades and texture as per customer requirement. Heat reflector units can be installed between the cladding if required by customer.

Weather seal systems

Weather seal systems are designed to provide all weather protection as well as to accommodate all deflections in building and door to ensure proper sealing. Sealing system are either coupled to aluminum extrusion or direct coupling arrangement as per requirement by customer. Track cleaners and flashing are provided at the bottom and top of door leaf respectively which act as both, track cleaners and weather seal at the same time. Moreover all the gaps are addressed to be finger-nip safe during operation. Friction pads are provided where possible friction between two parts may occur, hence protecting the main structure from any unwanted marks





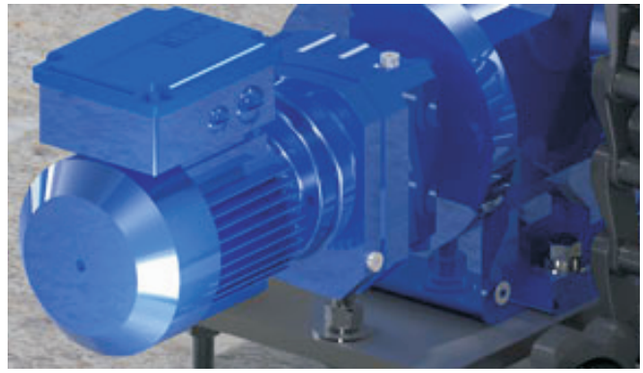
Buffers

Heavy duty shock absorbing buffers are provided at both side of door leafs which protect it from colliding from each other. These buffers are heavy duty energy absorbing to limit any direct shock to door or its driving system. Stoppers are provided at the end of each track to limit it from over running during manual towing operation.

Drive Units

Each door leaf is driven by its own motorized driving unit to ensure power operation. Driving unit used is of a reputed international manufacturer and is totally enclosed, fan-cooled and is designed to with-stand the harshest environments and heavy usage. Integrated brake system is incorporated for the purpose of emergency braking.It is coupled with high efficiency reducing gearbox with greatly oversized output bearings for extended life and greater reserve load capacity. Case design limits it for any possible oil leaks. Center bore stub axle allows maintenance crews to pressure lubricate the sealed automotive bearings through a built in zerk fitting.

There is no need to disassemble the unit to repack the wheel bearings. Weather hood protects the motor assembly on exterior mounts or during wash down for interior applications. Hood also acts as a safety guard for drive components. The hood is hinged for interior applications and bolted for exterior mount. Power supply to door are provided by overhead bus bars and are capable of operating effectively in maximum anticipated temperatures.



Variable Frequency Drive



Driving unit is provided with the variable frequency drive which allows door to gradually accelerate and decelerate in normal working conditions. VFD mechanism limit the door of any immediate shocks and inclination. The unit is field programmed to adapt to a specific requirement or owner preference.

Programmable Parameters:

1 Torque boost: To get even the largest and heaviest doors moving.

2 Acceleration Time: Smoothly gets the door started.

No more sudden jerky movements or tire spin on start up.

3 Deceleration Time: Brings door to a slow and smooth stop. No more violent stops from full speed.

4 Top Speed: Allows field adjustment to the projects needs.

5 Multiple Speeds: Allows door to operate in high speed mode in wide openings and then slow down to normal speed for parking.

6 Built in Diagnostics: Alerts technician to problems with building power such as low voltage or power spikes.

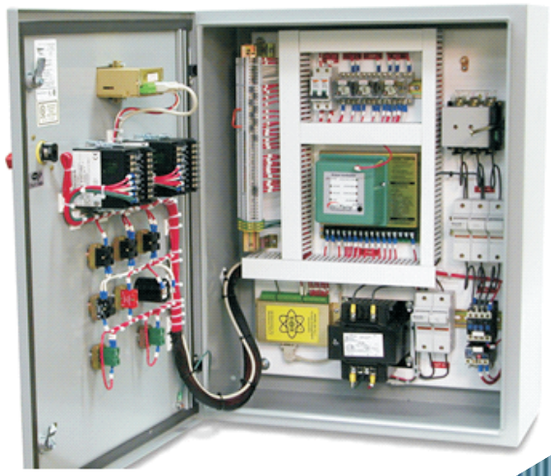


Safety Devices.

Door operating is integrated by safety devices which shall avoid the possible accidents. Limit switches limits doors from the unplanned travel. Photo censors are installed on the leading edge of the door leaf to detect unwanted presence and stop its operating immediately. Visual and audible warning are incorporated in safety system which operates during working of gate.

Control Panel

Control Panel is enclosed in a powder coated cabinet which is mounted into the bottom section of the door leaf. The construction is compact to accommodate itself within the limited space but with the ease of access.It is programmed for efficient working of door leaf without any possible breakdowns and is capable of controlling every possible aspect of operation including safety devices. Operating pendant enables to operate the door easily and is fitted on the leading edge of the door for safety concerns. It can be provided with the cord of desired





NIHVA

We Make Possible

Automation Solutions	Material Handling Equipments	Door Automation
Pick & Place Machines	Dock Levellers	High speed doors / Sectional doors
Manipulator /Air Balancer	Scissor lifts / Mobile ramp	Sliding doors / Folding Doors
Wireless Automation (M2M)	Dock Shelters / Dock Seals / Dock House	Fire rated - Mot. Rolling shutters / Sliding doors
SPM	Goods Lift	Hanger Doors / Extra large opening doors
Automation Softwares	Spiral Conveyors / Belt Conveyors	Boom barriers / Turnstiles / Bollards
Grease / Oil Separator	Telescopic Conveyors	Sliding / Cantilever Gates

International Business Partners

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